Aircraft Wing Lift Calculator Engineering Notebook

Week 1

Date: September 2026

Project Goal

I wanted to understand how aircraft wings generate lift and why airplanes are able to stay in the air despite their weight.

What I Worked On

- Researched the basic principles of aerodynamics.
- Studied the lift equation and its variables.
- Learned about air density, velocity, wing area, and lift coefficient.
- Created the GitHub repository and project outline.
- Designed the project visual and README structure.

What I Learned

I learned that lift depends on several factors working together rather than a single force. Small changes in speed or wing characteristics can significantly affect aircraft performance.

Challenges

At first, I found it difficult to understand the lift coefficient and how engineers estimate its value in real aircraft.

Results

- Project structure completed.
- Research phase completed.
- First version of the lift equation implemented in Python.

Next Steps

- Complete the working calculator.
- Test different flight conditions.
- Verify calculations with sample values
-

Week 2

Date: September 2026

What I Worked On

- Completed the first working version of the Aircraft Wing Lift Calculator.
- Added user inputs for:
    - Air Density
    - Velocity
    - Wing Area
    - Lift Coefficient
- Calculated lift force using the standard lift equation.
- Improved project documentation.
- Finalized project visuals for GitHub.

What I Learned

Working on this project helped me understand how engineers combine mathematics, physics, and programming to solve real-world problems.

I also learned that the same aircraft can generate very different amounts of lift depending on altitude, air density, and flight speed.

Results

The calculator successfully calculated lift force using different flight conditions and input values.

Example

- Air Density = 1.225 kg/m^3
- Velocity = 120 m/s
- Wing Area = 30 m^2
- Lift Coefficient = 0.85

Output

Lift Force ≈ 224,910 N

Challenges

One of the parts I struggled with at first was understanding how aerodynamic concepts are transformed into mathematical equations that can be used in a computer program.

Key Lessons Learned

- Lift is generated through the interaction of multiple variables.
- Mathematics provides the foundation for engineering analysis.
- Programming can be used to model and visualize engineering concepts.
- Engineering is about understanding relationships between variables and predicting outcomes.

Final Reflection

This project helped me better understand one of the most fundamental questions in aviation:

How does an aircraft stay in the air?

Before starting this project, I understood the basic concept of lift, but I had never explored the factors behind it in detail.

Building a simple calculator allowed me to connect physics, mathematics, and programming in a practical way.

The most rewarding part was seeing how a formula I had only read about could become a working tool that I built myself using Python.

Flight begins long before takeoff. It begins with understanding the science behind it.

Next Steps

- Complete final project documentation.
- Prepare engineering notebook and project report.
- Validate calculations using sample data.
- Publish final project materials on GitHub.

Engineering Insight

The results demonstrated that velocity has the greatest impact on lift generation because the lift equation contains a velocity-squared term. This means relatively small increases in airspeed can produce substantial increases in lift force.

Week 3

Date: September 2026

What I Worked On

- Completed the Aircraft Wing Lift Calculator project.
- Finalized project documentation and README.
- Reviewed and validated lift calculations.
- Organized engineering notes and project reports.
- Improved project presentation for GitHub.
- Prepared project materials for portfolio and LinkedIn publication.

What I Learned

This final stage helped me understand the complete engineering workflow from research and problem definition to implementation and documentation.

I learned that engineering projects require more than calculations. Clear documentation, validation, communication, and presentation are also essential parts of the engineering process.

Results

✅ Aircraft Wing Lift Calculator completed.

✅ Engineering notebook completed.

✅ Project report completed.

✅ GitHub repository finalized.

✅ Aerodynamic visualization created.

✅ Lift calculations successfully validated using sample flight conditions.

Challenges

One challenge was transforming aerodynamic theory into a practical engineering tool while keeping the project simple and understandable.

Another challenge was documenting the project in a professional engineering format.

Key Lessons Learned

- Lift generation depends on the interaction of multiple aerodynamic variables.
- Engineering equations can be transformed into practical software tools.
- Programming is a powerful method for visualizing scientific concepts.
- Documentation is an important component of engineering projects.
- Small projects can be valuable learning experiences when approached systematically.

Final Reflection

This project began with a simple question:

How does an aircraft generate lift?

By researching aerodynamics, studying the lift equation, and building a working calculator in Python, I gained a much deeper understanding of the science behind flight.

The most rewarding part of the project was seeing a theoretical aerospace engineering concept transformed into a practical software tool that I built myself.

This project strengthened my skills in Python programming, mathematical modeling, problem solving, and aerospace engineering fundamentals.

Next Steps

- Develop Aircraft Wing Lift Calculator V2.
- Add drag force calculations.
- Implement lift-to-drag ratio analysis.
- Explore stall prediction techniques.
- Build an interactive aerodynamic dashboard.

Project Status

✅ Completed

Final Project Deliverables

- Python Aircraft Wing Lift Calculator
- Engineering Notebook
- Engineering Project Report
- Aerodynamic Visualization
- GitHub Repository Documentation

Closing Statement

This project successfully combined aerospace engineering fundamentals, physics, mathematics, and software development to model a key aviation concept. It demonstrates how engineering principles can be transformed into practical computational tools and represents an important step toward more advanced aerospace engineering projects.

Flight begins long before takeoff. It begins with understanding the science behind it.

Sarp Akar

Oct 2026